BUS13-3

Financial Accounting & Statements

Reading the three statements as constructed documents — ratios, working capital, and earnings quality

Accounting tells you what happened; finance asks what it is worth and whether you can trust the numbers.

Albert School — 22 hours

Contents

Preface

This is the entry point to the Finance arc. You arrive with no accounting prerequisites, and that is deliberate: we teach the financial statements from the **user's** perspective — the analyst, the investor, the manager who must act on the numbers — not from the bookkeeper's. We will not spend time on debits and credits or on how a ledger is posted. Instead, we treat the balance sheet, the income statement and the cash flow statement as finished documents that someone **built**, shaped by accounting rules and by the incentives of the people who prepared them, and we learn to read them critically.

Everything introduced here is assumed by the courses that follow. In Semester 2, **Financial Diagnosis & Management Control** will turn the ratios and working capital you compute here into a full diagnosis. **Corporate Finance** (S3) and **Valuation** (S4) will take the cash flows you learn to extract and discount them into a value. **Corporate Finance Leadership** (S6) will assume all of it as second nature. So the goal of this course is not encyclopaedic coverage — it is fluency: the ability to open a real annual report, find the numbers that matter, compute what they imply, and ask the right sceptical questions.

A word on method. Each idea is introduced with a concrete example first and the general principle second, because that is the order in which understanding actually forms. A single fictional company, **Lumio** (a mid-size lighting manufacturer), runs through the whole book so that every statement, ratio and adjustment refers back to one coherent set of numbers.

1 What accounting measures, and what finance cares about

Start with a puzzle. Lumio's balance sheet says the company is "worth" €350 million — that is its **book equity**, assets minus liabilities. Yet on the same day its shares trade on the market for a total of €900 million. Both numbers are correct. They simply answer different questions. Understanding why is the whole foundation of this course.

Definition 1 **Accounting** is the system of rules for recording, classifying and reporting a firm's transactions in monetary terms. Its output is the set of **financial statements**. **Finance** is the discipline of decision-making about value, risk and cash over time. Accounting tells you what **has happened** in the language of recorded transactions; finance asks what the firm is **worth** and what it **should do next**.

Accounting is built to be reliable, auditable and comparable. To achieve that it makes conservative conventions — chiefly, recording most assets at what was actually **paid** for them. Finance cares about the future: expected cash flows, their risk, and their timing. The two diverge whenever the past paid-for amount is a poor guide to future value.

Worked example — three places where book value misleads.

1. **A brand.** Lumio has spent decades building its brand, but you cannot "buy" a brand in a single transaction, so accounting records it at zero. The brand is plainly worth something — it just never appears.
2. **Land bought in 1975.** A warehouse plot carried at its 1975 purchase price of €2 m is worth €40 m today. The balance sheet still says €2 m.
3. **A loss-making patent on the books at cost.** A patent capitalised at €30 m has been made obsolete by a competitor; its true economic value is near zero, but until an impairment is booked it sits at €30 m.

These are not accounting **errors**. They follow directly from the conventions — and that is exactly why the analyst must read past the recorded number.

Definition 2 Under the **historical cost** convention, an asset is recorded at the amount paid to acquire it, less accumulated depreciation, and is generally not written up when its market value rises. Under **fair value**, an asset is carried at the price it would fetch in an orderly market today. Most operating assets are held at historical cost; many financial instruments are held at fair value.

Common pitfall "The balance sheet shows what the company is worth." It does not. It shows the recorded cost of what the company **owns and owes** under a specific set of conventions. Market value, replacement value and book value are three different numbers, and confusing them is the single most common beginner's error in this course.

The reconciling discipline throughout is to ask, of any line: **is this a paid-for past amount or an estimate of future value, and which one does my decision need?**

2 The balance sheet: a snapshot of what is owned and owed

A balance sheet answers one question at one instant: what does the firm control, and who has a claim on it? Everything the firm controls (its **assets**) was financed by someone — either by lenders and suppliers (**liabilities**) or by owners (**equity**). Hence the identity that the document can never violate.

Definition 3 The **accounting equation**:

Assets = Liabilities + Equity.

Equivalently, **equity** — the owners' residual claim — is what is left of the assets after every creditor is paid: Equity = Assets − Liabilities.

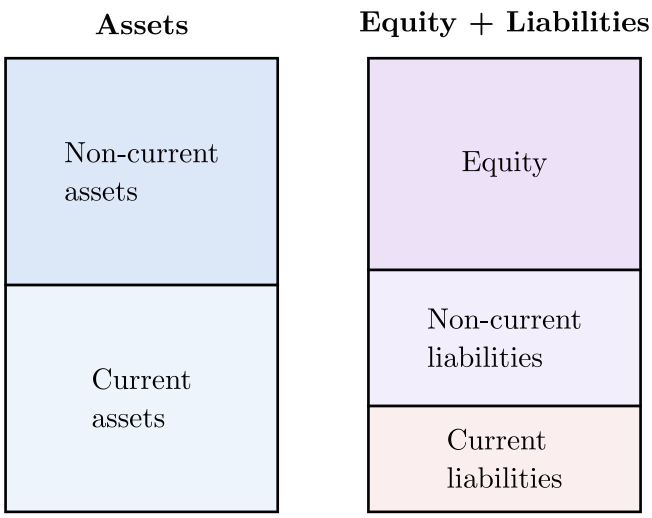

Figure 1: The two sides of the balance sheet always sum to the same total: every asset is financed by a claim. Assets are ordered by how soon they turn to cash; claims by how soon they fall due.

2.1 Reading the asset side

Definition 4 An asset is **current** if it is expected to be converted to cash, sold or consumed within one year (or the firm's operating cycle); it is **non-current** otherwise. Typical current assets: cash, marketable securities, **receivables** (amounts customers owe), **inventory**. Typical non-current assets: **property, plant & equipment** (PP&E), and **intangibles** (patents, capitalised software, goodwill).

The ordering is not cosmetic: it tells you, at a glance, how much of what the firm controls could be turned into cash quickly — the raw material for the liquidity ratios of Chapter 8.

2.2 Reading the claims side

Definition 5 **Liabilities** are obligations to outside parties: **payables** (amounts owed to suppliers), debt, accrued expenses, deferred revenue. They are split **current** (due within a year) and **non-current**. **Equity** is the owners' residual claim: capital contributed by shareholders plus **retained earnings** — the cumulative profit the firm has kept rather than paid out as dividends.

Worked example — Lumio's balance sheet, €m.

Assets		**Equity & liabilities**	
PP&E	400	Share capital + reserves	305
Intangibles	100	Retained earnings	45
Non-current assets	**500**	*Equity*	**350**

Assets		Equity & liabilities	
Inventory	120	Long-term debt	200
Receivables	90	*Non-current liabilities*	**200**
Cash	40	Payables	80
		Short-term debt	60
		Accruals	60
Current assets	**250**	*Current liabilities*	**200**
Total assets	**750**	**Total equity & liab.**	**750**

2.3 Why book equity is not market equity

Lumio's book equity is €350 m. Three forces drive its €900 m market value above that figure: unrecorded intangibles (the brand, the workforce), assets carried below current value (the old land), and — above all — **expected future profits**, which the balance sheet does not record at all. Book equity looks backward at recorded cost; market equity looks forward at expected cash flows discounted to today.

Common pitfall A company can have **negative** book equity and still be perfectly healthy — think of a mature firm that has bought back most of its shares, or a franchise whose brand (its real asset) is unrecorded. Negative book equity is a flag to investigate, not a verdict.

3 The income statement: performance over a period

Where the balance sheet is a photograph at an instant, the income statement is a film of one period (a quarter, a year). It starts from what the firm earned and subtracts, in a deliberate order, the costs of earning it — ending at the profit left for owners.

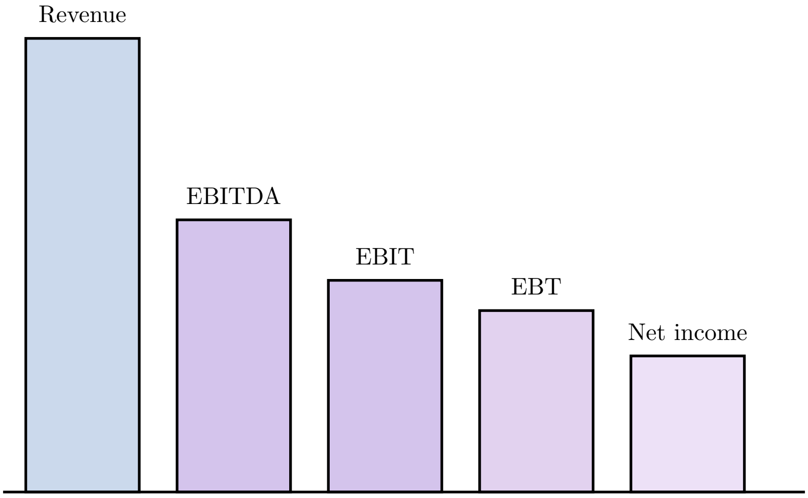

Figure 2: The margin cascade. Each step strips out a further layer of cost: from Revenue to EBITDA (operating costs except depreciation), to EBIT (after depreciation), to EBT (after interest), to Net income (after tax). Heights are illustrative of Lumio's figures.

Definition 6 Reading down the income statement:

- **Revenue** (sales): the value of goods and services delivered in the period.

- **Gross profit** = Revenue − **cost of goods sold** (COGS, the direct cost of what was sold).
- **EBITDA** = earnings before interest, taxes, depreciation and amortisation — gross profit less other operating costs (SG&A), but before the non-cash depreciation charge. A proxy for operating cash generation.
- **EBIT** (operating profit) = EBITDA − **depreciation & amortisation** (D&A).
- **EBT** (pre-tax profit) = EBIT − net interest and other financing costs.
- **Net income** = EBT − tax. The "bottom line": profit attributable to owners.

3.1 Operating versus financing

The single most important cut on the income statement is between **operating** items (how well the business runs) and **financing** items (how it is paid for). EBIT is the dividing line: everything above it describes operations; interest, below it, describes the capital structure.

> **Worked example — why the cut matters.** Two lighting makers each earn EBIT of €80 m. Firm A has no debt; Firm B borrowed heavily and pays €60 m of interest. Their operations are identical — same EBIT — but B's net income is far lower. Judge the **business** on EBIT/EBITDA; judge the **financing decision** separately. Mixing the two is how a good business with bad financing gets mistaken for a bad business.

Definition 7 A **margin** is a profit measure divided by revenue. **Gross margin** = gross profit / revenue measures pricing power and production efficiency; **EBIT (or operating) margin** measures operational profitability; **net margin** measures what finally reaches owners per euro of sales.

For Lumio: gross margin = 240/600 = 40%; EBITDA margin = 120/600 = 20%; EBIT margin = 80/600 ≈ 13.3%; net margin = 45/600 = 7.5%.

Common pitfall EBITDA is popular because it strips out financing, tax and accounting depreciation choices — but for that same reason it **flatters** capital-intensive firms. Depreciation is a real economic cost: the machines wear out and must be replaced. A company that lives on "EBITDA" while quietly under-investing is borrowing from its own future. Never let EBITDA make you forget capex.

4 Revenue recognition: when is a sale a sale?

The number at the very top of the income statement — revenue — looks the most solid and is in fact one of the most manipulable. The question is timing: in **which period** should a sale be recorded?

> **Worked example — the gym membership.** A customer pays Lumio's services arm €1,200 on 1 January for a 12-month maintenance contract. It is tempting to book €1,200 of revenue in January. But Lumio has not yet **done** anything — it owes eleven more months of service. The correct treatment records €100 of revenue each month as the service is delivered; on 1 January the €1,200 is **deferred revenue**, a liability (an obligation to perform), not revenue.

Definition 8 The **revenue recognition principle**: revenue is recognised when the firm has **transferred the promised goods or services** to the customer — i.e. when it has performed — not necessarily when cash changes hands. Cash received before performance is **deferred (unearned) revenue**, a liability; performance completed before cash is received creates a **receivable**.

Because recognition is a matter of judgement, it is a favourite lever for inflating reported earnings. Three classic patterns:

Definition 9

- **Channel stuffing**: shipping more product to distributors than they can sell, booking it as revenue now and quietly accepting returns later.
- **Long-term contracts**: on multi-year contracts, recognising revenue by **percentage of completion** invites optimistic estimates of how "complete" a project is, pulling tomorrow's revenue into today.
- **Subscription timing**: recognising the full annual fee up front rather than spreading it over the service period.

Common pitfall When revenue grows much faster than cash collected from customers — receivables and "unbilled revenue" ballooning — be suspicious. Aggressive recognition shows up as revenue that has not yet (and may never) turn into cash.

5 The cash flow statement: following the money

Lumio reported €45 m of net income. Did its bank balance rise by €45 m? No — it **fell** by €7 m. Profit and cash are not the same thing, and the gap between them is where many companies quietly die. The cash flow statement exists to explain that gap.

Definition 10 The **cash flow statement** reconciles the change in the firm's cash over a period, split into three activities:

- **Operating** (CFO): cash generated by the core business.
- **Investing** (CFI): cash spent on / received from long-term assets — chiefly **capital expenditure** (capex).
- **Financing** (CFF): cash from / to providers of capital — debt drawn or repaid, equity issued, dividends paid.

The three sum to the net change in cash.

5.1 The indirect method

Net income and operating cash flow differ for two reasons: the income statement contains **non-cash** charges (like depreciation), and it records revenues and expenses on an **accrual** basis that does not match the timing of cash. The indirect method starts from net income and undoes both.

Definition 11 **Operating cash flow, indirect method**: start from net income; **add back** non-cash charges (depreciation, amortisation); then adjust for **changes in working**

capital — subtract increases in receivables and inventory (cash tied up), add increases in payables (cash deferred).

Worked example — Lumio's operating cash flow, €m.

Net income	45
+ Depreciation & amortisation	40
− Increase in receivables	(10)
− Increase in inventory	(15)
+ Increase in payables	8
Operating cash flow (CFO)	**68**
− Capital expenditure (investing)	(60)
− Dividends + net debt (financing)	(15)
Net change in cash	**(7)**

Depreciation is added back because it reduced profit without any cash leaving. The €10 m rise in receivables is sales booked but not yet collected — profit without cash. The result: €45 m of profit, but cash **fell** by €7 m.

Common pitfall A profitable company can run out of cash. Fast-growing firms are most at risk: every new sale ties up more cash in inventory and receivables **before** the customer pays. "Profitable but cash-negative" is not a contradiction — it is the normal signature of growth, and the reason CFO is watched as closely as net income.

6 How the three statements connect

The statements are not three independent reports; they are three views of one reality, locked together. Master the linkages and you can trace any transaction end to end — and spot when a set of statements does not hang together.

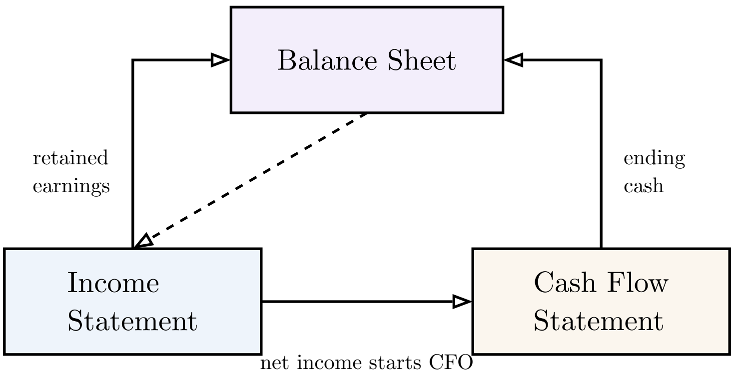

Figure 3: The three-statement loop. Net income flows into operating cash flow and (less dividends) into retained earnings; ending cash from the cash flow statement becomes the cash line on the next balance sheet; and the balance sheet's assets and debt feed back as depreciation and interest on the next income statement.

Definition 12 The three core linkages:

1. **Net income → equity.** Net income, less dividends, increases retained earnings on the balance sheet.

2. **Net income → cash.** Net income is the starting point of operating cash flow; the statement's ending cash becomes the balance sheet's cash line.
3. **Capex → asset base → depreciation.** Investing cash flow builds PP&E; that PP&E is then depreciated, a charge on future income statements. Likewise, financing decisions set the debt whose interest hits future income statements.

Worked example — tracing one transaction. Lumio sells €100 of lamps on credit, costing €60 to make. Trace it:

- **Income statement**: revenue +100, COGS +60, so profit +40 (before tax).
- **Balance sheet**: inventory −60 (lamps shipped), receivables +100 (customer owes), retained earnings +40. The equation still balances.
- **Cash flow**: so far, **zero** cash. CFO shows +40 net income but −100 receivables and +60 inventory release → net €0. Only when the customer pays does €100 of cash arrive (receivables −100, cash +100).

One sale; three coordinated effects; cash arriving last.

7 Earnings quality: can you trust the bottom line?

Two firms each report €45 m of net income. In one, that €45 m is the repeatable output of normal operations. In the other, it includes a €30 m one-off gain from selling a building and would have been €15 m otherwise. The reported numbers are equal; their **quality** is not.

Definition 13 **Earnings quality** is the degree to which reported earnings reflect sustainable, repeatable operating performance rather than one-off events or accounting choices. **Normalized (or underlying) earnings** are reported earnings adjusted to remove **non-recurring items** — gains or charges not expected to repeat.

Worked example — normalising Lumio. Suppose Lumio's €45 m net income includes a €12 m gain from selling a warehouse and a €5 m restructuring charge (both pre-tax, tax 25%). Normalised earnings remove the gain and add back the charge:

$$45 - 12 \times 0.75 + 5 \times 0.75 = 45 - 9 + 3.75 = 39.75 \text{ m.}$$

The €39.75 m is the better basis for forecasting next year — and for valuation.

Common pitfall **Red flags for low earnings quality** (each is a question, not a conviction):

- net income rising while operating cash flow stagnates or falls;
- receivables or inventory growing much faster than sales;
- recurring "one-off" charges that appear every single year;
- profits propped up by frequent asset sales rather than operations;
- a sudden change in an accounting estimate (useful life, bad-debt allowance) that conveniently boosts earnings.

The discipline: compare net income with CFO over several years. Quality earnings are eventually confirmed by cash.

8 Financial ratios: the four families

A single number in isolation says little — €45 m of profit is impressive for a corner shop and catastrophic for an oil major. Ratios put numbers in context by relating them to each other. There are four families, each answering a different question about the firm's health. All figures below use Lumio's statements.

Definition 14 **Liquidity** — can the firm pay its short-term bills?

$$\text{Current ratio} = \frac{\text{current assets}}{\text{current liabilities}}, \quad \text{Quick ratio} = \frac{\text{current assets} - \text{inventory}}{\text{current liabilities}}.$$

Lumio: current $= 250/200 = 1.25$; quick $= 130/200 = 0.65$. The quick ratio excludes inventory because inventory is the hardest current asset to turn into cash fast.

Definition 15 **Solvency** — can the firm survive its long-term debt load?

$$\text{Debt-to-equity} = \frac{\text{total debt}}{\text{equity}}.$$

Lumio: $(200 + 60)/350 = 260/350 \approx 0.74$. Higher means more leverage — more risk, but potentially more return to equity.

Definition 16 **Profitability** — how much profit per euro of sales (and of capital)?

$$\text{Gross margin} = \frac{\text{gross profit}}{\text{revenue}}, \quad \text{Net margin} = \frac{\text{net income}}{\text{revenue}}, \quad \text{ROE} = \frac{\text{net income}}{\text{equity}}.$$

Lumio: gross $= 40\%$; net $= 7.5\%$; ROE $= 45/350 \approx 12.9\%$.

Definition 17 **Efficiency** — how hard does the firm work its assets?

$$\text{Asset turnover} = \frac{\text{revenue}}{\text{total assets}}.$$

Lumio: $600/750 = 0.80$ — each euro of assets generates €0.80 of sales per year.

Common pitfall A ratio is meaningful only **in comparison** — against the same firm over time (trend) or against peers in the same industry (benchmark). A net margin of 7.5% is excellent for a grocer and poor for a software firm. Never interpret a ratio in a vacuum, and always check that numerator and denominator are measured consistently (e.g. total debt vs total liabilities for debt-to-equity — define your terms before you compare).

9 Working Capital Requirement (BFR)

We saw that growth ties up cash in receivables and inventory before customers pay. The **Working Capital Requirement** — in French, **Besoin en Fonds de Roulement** (BFR) — measures exactly how much cash the operating cycle locks up.

Worked example — the lag that costs cash. Lumio buys components, holds them as inventory, builds and ships lamps, then waits for customers to pay — meanwhile it owes its own suppliers. If it holds €120 m of inventory and is owed €90 m by customers, but owes only €80 m to suppliers, then €130 m of cash is permanently tied up in simply **running** the business, before a single euro of profit or investment.

Definition 18 The **Working Capital Requirement** (BFR) is the operating capital the business must finance:

$$\text{BFR} = \text{Inventory} + \text{Receivables} - \text{Payables}.$$

It is the cash absorbed by the gap between paying suppliers and collecting from customers. Lumio: $120 + 90 - 80 = 130$ m.

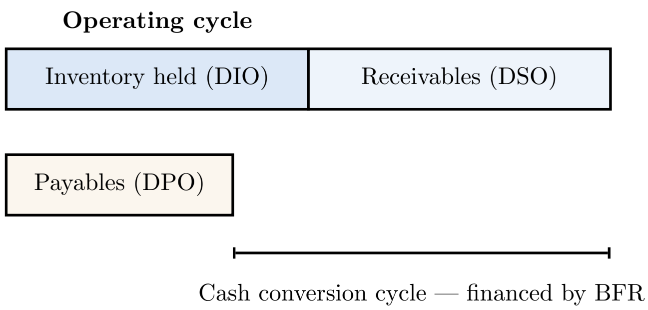

Figure 4: The operating cycle. Cash leaves when suppliers are paid and returns only when customers pay. The shaded gap — inventory and receivable days, less the days suppliers finance — is the cash conversion cycle that BFR funds.

Definition 19 The drivers of BFR, expressed in days:

- **DIO** (days inventory outstanding): how long stock sits before sale.
- **DSO** (days sales outstanding): how long customers take to pay.
- **DPO** (days payable outstanding): how long the firm takes to pay suppliers.

The **cash conversion cycle** $= \text{DIO} + \text{DSO} - \text{DPO}$. The longer it is, the more cash the operating cycle consumes.

Common pitfall BFR can be **negative**, and that is a powerful position, not a problem. Supermarkets collect from customers instantly (cash at the till, DSO ≈ 0) while paying suppliers weeks later (high DPO) — so suppliers finance their inventory. Negative BFR means the operating cycle **releases** cash as the firm grows.

10 Cash flow patterns across the company lifecycle

The **signs** of the three cash flows — operating, investing, financing — together tell a story about where a company is in its life. Read them as a pattern.

Definition 20 Typical lifecycle signatures (sign of CFO / CFI / CFF):

Stage	CFO	CFI	CFF	Reading
Startup	−	−	+	Burning cash, investing, funded by raised capital
Growth	+	−	+	Operations turn positive but still raising to invest
Mature	+	−	−	Self-funds investment and returns cash to investors
Decline	+	+	−	Selling assets; CFO may weaken; repaying/returning

Worked example — diagnosing from signs alone. A firm shows CFO +€68 m, CFI −€60 m, CFF −€15 m (Lumio's pattern). Operations generate cash, most of which is reinvested in capex, and the remainder returns to investors as dividends and debt repayment. This is the signature of a **mature, healthy** company — and it matches everything else we know about Lumio.

11 French GAAP vs IFRS

The same company can report different numbers depending on the rulebook. **IFRS** (International Financial Reporting Standards) is used by listed groups across the EU and much of the world; **French GAAP** (the Plan Comptable Général) governs the individual statutory accounts of French entities. For cross-border comparison, the differences matter.

Definition 21 Key differences in emphasis:

- **Philosophy**: IFRS is **principles-based** and leans toward **fair value** and "substance over form"; French GAAP is more **rules-based**, **prudent**, and anchored in **historical cost**.
- **Leases**: IFRS 16 puts almost all leases on the balance sheet (right-of-use asset + lease liability); French GAAP often keeps operating leases off it.
- **Development costs**: IFRS **requires** capitalising development costs that meet criteria; French statutory practice more often expenses them.
- **Revaluation**: IFRS permits revaluing certain assets to fair value; French GAAP generally does not.

Common pitfall Because IFRS capitalises leases and (some) development costs, an IFRS balance sheet will typically show **more assets and more debt** than a French-GAAP one for the very same business. When comparing a company reporting under IFRS to a peer under French GAAP, you are not comparing like with like until you adjust — leverage and asset-turnover ratios are especially sensitive.

12 Consolidated accounts

A group is many legal companies; investors want to see them as one economic entity. **Consolidation** combines a parent and the companies it influences into a single set of statements. The method depends on **how much** of the subsidiary the parent controls.

Definition 22 The three consolidation methods, by degree of control:

- **Full consolidation** (control, typically >50%): add 100% of the subsidiary's assets, liabilities, revenue and costs, then show the slice owned by outsiders as **non-controlling (minority) interests**.
- **Equity method** (significant influence, typically 20–50%): do not add the subsidiary's lines; instead carry the investment at cost plus the parent's share of its accumulated profits, shown on one line.
- **Proportional consolidation** (joint control): add the parent's **share** of each of the subsidiary's lines. (Now restricted under IFRS, but part of the historical toolkit and still seen.)

Worked example — same 30% stake, two stories. Lumio owns 30% of a components maker. Under the **equity method**, only Lumio's share of that company's profit appears, on one line — its revenue and debt stay off Lumio's statements. Had Lumio instead **controlled** it (full consolidation), 100% of the components maker's revenue and debt would appear, inflating both Lumio's apparent size and its leverage. Always check how a group consolidates before comparing its ratios with another's.

13 Capstone: reading a real annual report end-to-end

Everything in this book converges on one skill: opening a real annual report and producing a structured first-pass analysis. The point is not a verdict — it is a disciplined map of what the numbers say and what they leave you wanting to ask.

Definition 23 A **first-pass financial analysis** of a company comprises:
1. **Business model** — what the firm sells, to whom, and how it makes money.
2. **Key ratios** — at least one from each family (liquidity, solvency, profitability, efficiency), with a trend or peer benchmark.
3. **Working capital** — BFR and the cash conversion cycle, and whether they are rising.
4. **Three financial signals** — concrete observations from the numbers (e.g. "CFO lags net income three years running").
5. **Three structural or governance questions** — what warrants deeper digging (e.g. "why do 'non-recurring' charges recur every year?").

Worked example — a first pass on Lumio. Model. Mid-size lighting manufacturer; sells to distributors and runs a maintenance-services arm.

Ratios. Current 1.25, quick 0.65 (thin liquidity); debt-to-equity 0.74 (moderate leverage); gross margin 40%, net margin 7.5%, ROE 12.9% (solid profitability); asset turnover 0.80.

Working capital. BFR €130 m — substantial; watch whether it grows faster than sales.

Three signals. (1) Cash fell €7 m despite €45 m profit — growth is absorbing cash. (2) Quick ratio 0.65 means without selling inventory, current liabilities exceed liquid assets. (3) Net income of €45 m includes a €12 m one-off gain; normalised earnings are €40 m.

Three questions. (1) Why is the cash conversion cycle so long — can DSO be cut? (2) Is the €200 m long-term debt due soon, and at what rate? (3) Why was a warehouse sold this year — operational need, or earnings management?

The thread of this book. Accounting builds three interlocking statements under conservative conventions; finance reads them for value and for cash. Profit is not cash (the cash flow statement explains the gap); book value is not market value (the balance sheet explains the gap); and reported earnings are not always **quality** earnings (normalisation and the CFO-vs-net-income check expose the gap). Ratios and BFR turn raw statements into a diagnosis — and that diagnosis is where every later course in the Finance arc begins.